

RITHIKA MURUGAN

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OBJECTIVE: Exploring the health sector through an MSPH lens, with a curiosity for how qualitative and quantitative methods shape data interpretation. Passionate at viewing data through multiple perspectives to uncover deeper insights and drive impactful health solutions.

EDUCATION:

MSPH in Health Systems (GPA - 3.75)

Aug 2024 - May 2026 (expected)

Johns Hopkins University - Bloomberg School of Public Health

(Relevant courses: Statistical methods in Public Health, Introduction to Data Management, Qualitative Data Analysis)

B.E in Biomedical Engineering (CGPA - 9.4)

Aug 2020 - April 2024

SRM Easwari Engineering College, Chennai

INTERNSHIPS AND RELEVANT EXPERIENCE:

Research Assistant – SEE Change initiative, JHU

Aug 2025 - Ongoing

- Overseeing comprehensive data management for a Uganda-based randomized controlled trial (RCT) on small- and micro-entrepreneurs, including cleaning, coding, and merging qualitative and quantitative datasets, maintaining data integrity, and preparing data for advanced statistical analysis.
- Conducting mixed-methods analysis to examine behavioral, economic, and psychosocial factors influencing entrepreneurial performance, synthesizing findings into technical reports and recommendations to guide evidence-based program design and scalable entrepreneurship support initiatives.

Research Assistant – Rishi Foundation, JHU

June 2025 - Ongoing

- Implementing and evaluating a mixed-methods intervention combining physical activity, strength training, and nutrition to improve physical fitness, mental health, and resilience among children and youth living with or affected by HIV in South India.
- Conducting post-intervention data collection using validated tools (PHQ-9, GAD-7, CYRM-R), physical assessments, surveys, and focus group discussions, ensuring culturally sensitive and ethical community engagement throughout.
- Analyzing these quantitative and qualitative data using R to assess changes in health outcomes, program adherence, and impact; synthesizing findings into a report with recommendations for scale-up.

Research Assistant – School of Medicine, JHU

Dec 2024 - May 2025

- Conducted in-depth qualitative data analysis of over 35 interview transcripts using MAXQDA to systematically identify patterns, relationships, and emerging themes in the data.
- Led a critical component of the IMPACT smoking cessation initiative, investigating the application and effectiveness of ERIC (Expert Recommendations for Implementing Change) strategies in real-world implementation.
- Combined interview findings with a systematic review of literature from PubMed, CINAHL, and Implementation Science journals to assess alignment between empirical evidence and practitioner experiences.

Data Summarization for WHO meeting

Oct 2024 - Nov 2024

- Conducted qualitative coding of open-ended survey responses on community members and health workers values/preferences for self-care menstrual interventions.
- Used Dedoose software to analyze and code qualitative data from over 150 responses provided by women from diverse global locations.

Hospital Training at VHS Hospital, Tharamani

June 2023

- Collaborated with the Head of the Oncology Department, gaining hands-on experience in clinical operations and conducted a comprehensive analysis of medical equipment used in patient treatment to assess efficiency and effectiveness.
- Proposed and implemented improvements to the department's data management system, enhancing workflow and health record-keeping processes.

Systematic Review on psychological counseling for couples with primary infertility during ART

July - Aug 2022

- Conducted a systematic review adhering to PRISMA flowchart guidelines for data collection and presentation.
- Acquired in-depth understanding of the physical and psychological impacts of infertility on individuals within the local community in India.

SKILLS:

- *Programming languages* – R Studio, Stata, Python
- *Data analysis tools* – MAXQDA, Dedoose, QGIS 3.24, MS Excel, SQL, Redcap

PROJECTS:

BIOBENCH - Final year project

April 2024

Designed and developed a website/application to support local clinics and healthcare providers in economically disadvantaged areas of Chennai. Successfully implemented this project, free of charge, in two local clinics.

Medicine ordering system using Labview software

Aug 2023

Synthesized a structured for virtual medicine ordering and discount application.

CONFERENCE:

BIOBENCH: A comprehensive digital ecosystem for streamlined patient data management

Conference: 4th International conference on advances in management and technology, 2024.

Designing and executing a medicine ordering system using LabVIEW Software – an objective approach

Conference: RIBE 2023- International Conference on Research and Innovation in Biomedical Engineering.

CERTIFICATIONS:

Implementation Science and Research Practice certificate, 2025.

Ongoing

Leadership and team effectiveness, IIT Roorkee, 2023.

Percentage obtained: 80%

Foundation Certificate in Palliative Care, IIT Kanpur, 2023.

Percentage obtained: 84%

ACTIVITIES AND AWARDS:

Rishi Children's Fund Orphans and Vulnerable Children (OVC) Travel Grant at JHU

Awarded to support an 8–16-week field immersion in Tamil Nadu and Karnataka, India, focusing on community health research for HIV children, adolescents, and young adults.

Joint treasurer (2022-2023)

Energy Club (Interdepartmental and technical organization).

Managed finances and published over 30 content pieces and coordinated large-scale events during the tenure.

Member of YRC

Youth Red Cross – A Unit of Indian Red Cross Society, 2020-2023.

Organized impactful community activities, including blood donation drives, beach and lake cleanups, first aid training, and cancer awareness campaigns.

ACHIEVEMENTS:

- *Tutored* 2 distinction and merit achieving candidates for Acoustic piano (Grade 3), 2025.
- *First class Distinction holder* - SRM Easwari engineering college, 2024.
- Winner - Greenathon, 2024 - Proposed ideas for sustainable initiatives aimed at creating a greener campus.
- *Distinction rank holder* under Trinity College, London for Acoustic piano, (Grade-8), Level 3, Sept 2021.
- *Merit rank holder* under Trinity College, London for Theory of Music, (Grade-7), Level 3, June 2019.
- Colt karate – Black belt champion (Dan 1), 2018.